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CSE A-2

Subject: ProgrammingwithJava

ASSIGNMENT No : 7

TITLE :ImplementaJavaprogram to demonstrate inheritance and interfaces using suitable real-worldexamples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

LearningObjectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming

```
interface Vehicle {
    void start();
}

class Car {
    void fuelType() {
        System.out.println(x: "Fuel Type : Petrol");
    }
}

class SportsCar extends Car implements Vehicle {

    public void start() {
        System.out.println(x: "Sports Car Started.");
    }

    void display() {
        System.out.println(x: "Model : Ferrari");
    }
}

public class Main {
    Run | Debug
    public static void main(String[] args) {

        SportsCar sc = new SportsCar();

        sc.display();
        sc.fuelType();    // Inherited method
        sc.start();       // Interface method
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp7> java Main.java
Model : Ferrari
Fuel Type : Petrol
Sports Car Started.
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp7> █
```

Exercise

1.Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
class Shape {
    void display() {
        System.out.println(x: "Created a new shape!");
    }
}

class Circle extends Shape {
    double area (double r) {
        return 3.14 * r * r;
    }
}

class Rectangle extends Shape {
    double area (double l, double b) {
        return l * b;
    }
}

public class Ex1 {
    Run | Debug
    public static void main (String[] args) {
        Circle c = new Circle();
        Rectangle r = new Rectangle();

        c.display();
        System.out.println("Area of Circle : " + c.area(r: 10));
        r.display();
        System.out.println("Area of Rectangle : " + r.area(l: 10, b: 10));
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp7\Exercise> java Ex1.java
Created a new shape!
Area of Circle : 314.0
Created a new shape!
Area of Rectangle : 100.0
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp7\Exercise> █
```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
interface Product {
    void displayCategory();
}

class ProductDetails {
    int id;
    String name;
    double price;

    ProductDetails(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    void displayDetails() {
        System.out.println("Product ID   : " + id);
        System.out.println("Product Name : " + name);
        System.out.println("Price      : " + price);
    }
}

class Electronic extends ProductDetails implements Product {
    Electronic(int id, String name, double price) {
        super(id, name, price);
    }

    public void displayCategory() {
        System.out.println("Category   : Electronic");
    }
}

class Clothing extends ProductDetails implements Product {
    Clothing(int id, String name, double price) {
        super(id, name, price);
    }

    public void displayCategory() {
        System.out.println("Category   : Clothing");
    }
}

class Grocery extends ProductDetails implements Product {
    Grocery(int id, String name, double price) {
        super(id, name, price);
    }
}
```

```

    }
    public void displayCategory() {
        System.out.println("Category : Grocery");
    }
}

public class Ex2 {
    public static void main(String[] args) {
        Electronic e = new Electronic(101, "Laptop", 65000);
        Clothing c = new Clothing(102, "T-Shirt", 999);
        Grocery g = new Grocery(103, "Rice", 1200);
        e.displayDetails();
        e.displayCategory();
        System.out.println();
        c.displayDetails();
        c.displayCategory();
        System.out.println();
        g.displayDetails();
        g.displayCategory();
    }
}

```

```

PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp7\Exercise> java Ex2.java
Product ID      : 101
Product Name    : Laptop
Price           : 65000.0
Category        : Electronic

Product ID      : 102
Product Name    : T-Shirt
Price           : 999.0
Category        : Clothing

Product ID      : 103
Product Name    : Rice
Price           : 1200.0
Category        : Grocery
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp7\Exercise>

```